

Compressive Sensing of Stepped-Frequency Radar Based on Transfer Learning

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Abstract—It usually suffers from long observing time and interference sensitivity when a radar transmits the high-range-resolution stepped-frequency chirp signal. Motivated by this, only partial pulses of the stepped-frequency chirp are utilized. For the obtained incomplete frequency data, a Bayesian model based on transfer learning is proposed to reconstruct the corresponding full-band frequency data. In the training phase, a complex beta process factor analysis (CBPFA) model is utilized to statistically model each aspect-frame from a set of given full-band frequency data, whose probability density function (pdf) can be learned from this CBPFA model. It is important to note that the numbers of factors and dictionaries are automatically learned from the data. The inference of CBPFA can be performed via the variational Bayesian (VB) method. In the reconstruction phase for the incomplete frequency data that “related” to the training samples, its corresponding full-band frequency data can be analytically reconstructed via the compressive sensing (CS) method and Bayesian criterion based on the transfer knowledge of the previous pdfs learned from the training phase. About the “relatedness” between each training frame and the incomplete test frequency data, we utilize the frame condition distribution of incomplete test frequency data to represent. The proposed method is validated on the measured high range resolution (HRR) data.

Index Terms—Bayesian, complex beta process factor analysis, compressive sensing, stepped-frequency, transfer learning, variational inference.

I. INTRODUCTION

HIGH range resolution has many advantages in radar system [1]. However, in order to achieve high range resolution, one must use very short pulses (correspond to wide-band signals) and consequently reduce the average transmitted power, and impose severe operating bandwidth requirements [2]. Stepped frequency waveforms (SFW) can achieve high range resolution without sharing the disadvantages of the wide-band signals. SFW are a series of narrow-band pulses whose carrier frequency increases in steps of constant frequency between consecutive pulses. Each pulse can have its own linear

frequency modulation (LFM), or other type of modulation, in this paper LFM is assumed, i.e. stepped-frequency chirp signal [2]. By utilizing the bandwidth synthetic technique, the bandwidths of the pulses of the stepped-frequency chirp signal can be synthesized to a large bandwidth, then, the high range resolution is realized and the high range resolution (HRR) data of targets are achieved [3].

Since stepped-frequency chirp signal comprises a series of narrow-band pulses, it usually takes a long time to observe all the pulses. However, in addition to transmitting a high range resolution signal, the multifunctional radars often have other critical activities, such as searching and target tracking. As a result, observing time for each target is usually limited [4]. Another disadvantage of stepped-frequency chirp is the sensitivity to inference [4], such as radio-frequency interference, self-induced interference, and artificial interference, etc. [5]. Therefore, partial pulses with these interferences may be unusable. To avoid the long observing time and inference sensitivity of the stepped-frequency chirp signal, only partial pulses of the stepped-frequency chirp are utilized. Then, the incomplete frequency data resulting from the vacancy in frequency bands is obtained, which can be used to recover the corresponding full-band frequency data. Thus, the full bandwidth frequency data can be obtained without inference in a shorter time.

Recently, an emerging technique in information theory and signal processing called compressive sensing (CS) [6], [7] has received significant attentions in many fields and has been applied successfully in different applications [8]. CS can obtain better performance than some other reconstruction techniques, e.g. linear prediction techniques with interpolation and extrapolation, and similar methods [4], [9]. CS theory attempts to achieve accurate signal reconstruction while sampling signals at a low sampling rate, typically far lower than the Nyquist/Shannon rate. Let $\mathbf{x} \in C^{L \times 1}$ represent the unknown original signal. Then the signal sampling model for CS can be expressed as:

$$\mathbf{y} = \Phi \mathbf{x} + \xi \quad (1)$$

where $\mathbf{y} \in C^{J \times 1}$ is the measurements, $\Phi \in C^{J \times L}$ is the measurement matrix, and we are interested in the case $J \ll L$, $\xi \in C^{J \times 1}$ represents the measurement noise. CS theory shows that if the signal $\mathbf{x}$ can be sparsely rendered in a dictionary $\mathbf{Q} \in C^{L \times E}$, i.e., $\mathbf{x} = \mathbf{Q}\mathbf{g}$, where $\mathbf{g} \in C^{E \times 1}$ represents the sparse scaling coefficients, it can be exactly recovered with high probability based on a relatively small set of under-sampled measurements $\mathbf{y}$. In traditional sparse coding tasks, the dictionary $\mathbf{Q}$ is usually fixed and given.

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There are three major classes of computational techniques for solving this sparse approximation problem [10], [11]: *Greedy pursuit*, which builds up an approximation one step at a time by making locally optimal choices at each step, e.g. matching pursuit (MP) and orthogonal matching pursuit (OMP); *Convex relaxation*, which replaces the combinatorial problem with a convex optimization problem and solves the convex program with algorithms that exploit the problem structure; *Bayesian framework*, which assumes a prior distribution for the unknown coefficients that favors sparsity and develops a maximum posterior estimator that incorporates the observation, e.g. Bayesian compressive sensing (BCS) [12].

By utilizing CS, it is easy to realize the reconstruction of the incomplete frequency data. For our problem, $\mathbf{y}$ represents the given incomplete frequency data, and $\mathbf{x}$ denotes the corresponding full-band frequency data to be reconstructed. Therefore, Φ is a matrix whose function is extracting partial elements from $\mathbf{x}$. Since the restricted isometry property (RIP) condition can be achieved with high probability via a random projection matrix, CS theory typically adopts random matrices as the perfect candidates for Φ [13]. However, we prefer Φ to have physical meaning in some applications, for example the proposed extracting matrix in this paper. In [14], Φ is also designed as the extracting matrix, although the RIP condition may not be satisfied, CS can still achieve good experimental results in the performance of the reconstruction. Therefore, we choose Φ as the extracting matrix in this paper.

Although the traditional CS methods can obtain better reconstruction performance, they realize the reconstruction by only utilizing the underlying sparse property of $\mathbf{x}$ and fixed dictionary $\mathbf{Q}$. To address this, we propose an alternative statistical compressive sensing algorithm based on transfer learning [15] in this paper. Our method utilizes a set of full-band frequency data as the auxiliary/training set, which can be partitioned into many aspect-frames due to the target-aspect sensitivity. For each aspect-frame, a complex beta process factor analysis (CBPFA) model is developed to automatically infer the number of factors and learn the dictionary. In the reconstruction stage, by utilizing the probability density function (pdf) of each aspect-frame learned from the CBPFA model as the transferred knowledge, the reconstruction of the out-of-sample data (i.e. incomplete frequency data) can be realized analytically by CS theory and Bayes' rule.

It is worth noting that no matter if the training aspect-frames and the incomplete frequency data $\mathbf{y}$ are with the same or different category, they should be "related", explicit or implicit, to each other at least [15]. If they are not related, transfer learning may degrade the reconstruction performance. However, how to assess the degree of relatedness is crucial. Motivated by this, we utilize the frame condition distribution of $\mathbf{y}$ as the relatedness between each training aspect-frame data and the incomplete frequency data $\mathbf{y}$, which provides the flexibility of no sharing when they are unrelated and partial/total sharing when they are related.

The remainder of the paper is organized as follows: in Section II, we propose a CBPFA model for each aspect-frame of the given training full-band data set and calculate the corresponding pdf. In Section III we introduce how to reconstruct the incomplete frequency data by utilizing/transferring the pdfs of the aspect-frames learned in the training stage.

Section IV presents the experimental results on measured HRR data. Section V concludes the paper and discusses the possible future work.

II. BAYESIAN STATISTIC LEARNING OF THE TRAINING FULL-BAND FREQUENCY DATA

A. Introduction of the Stepped-Frequency Chirp Signal

The stepped-frequency chirp signal is a sequence of narrow-band chirp pulses, whose carrier frequency from pulse to pulse is stepped by a fixed frequency step [3], [16]. Suppose the wide-band/synthetic signal with carrier frequency f_c , bandwidth B_p , chirp rate γ , and pulse length T_p ; the narrow-band chirp signals with bandwidth B_H and pulse length T_{pH} ; the frequency step is f_{step} , and the number of pulses is H . Therefore, the carrier frequency of the h th narrow-band pulse is given by:

$$f_c(h) = f_c + \left(h - \frac{1}{2} - \frac{H}{2}\right) f_{step} \quad (2)$$

where $h \in \{1, 2, \dots, H\}$. The phase of the transmitted pulses belonging to one burst can be described by:

$$\mathbf{s}_h(t) = \text{rect}\left(\frac{t}{T_{pH}}\right) \exp(j2\pi f_c(h)t) \cdot \exp(j\pi\gamma t^2) \quad (3)$$

The signal phase received by the radar system from a single scatterer at a distance R is therefore given by:

$$\mathbf{s}_h(t) = \text{rect}\left(\frac{t - \tau}{T_{pH}}\right) \exp(j2\pi f_c(h)(t - \tau)) \cdot \exp(j\pi\gamma(t - \tau)^2) \quad (4)$$

where $\tau = 2R/c$ represents the time delay of the echo with c is the speed of light. The appropriate reference function for the demodulating and motion compensating the received signal is:

$$\mathbf{s}_{ref_h}(t) = \exp(j2\pi f_c(h)t) \quad (5)$$

The signal that results from mixing the received signal of (4) and the complex conjugate of the reference function of (5) is:

$$\mathbf{s}_h(t) = \text{rect}\left(\frac{t - \tau}{T_{pH}}\right) \exp(-j2\pi f_c(h)\tau) \cdot \exp(j\pi\gamma(t - \tau)^2) \quad (6)$$

By means of the pulse compression of $\mathbf{s}_h(t)$, we can obtain the frequency domain signal:

$$\mathbf{s}_h(f) = \text{rect}\left(\frac{f}{B_H}\right) \exp(-j2\pi f\tau) \exp(-j2\pi f_c(h)\tau) \quad (7)$$

An important technique for adopting the stepped-frequency chirp signal is the bandwidth synthetic technique. It can realize that the high range resolution by recombining the narrow-band pulses to obtain the wide-band pulse. For example, the commonly used frequency domain bandwidth synthetic method [3]. It shifts the frequency spectrums $\{\mathbf{s}_h(f)\}_{h=1}^H$ of the pulses in frequency and adds to a large effective bandwidth pulse, to get the full-band frequency data (i.e. the frequency spectrum of HRR data) $\mathbf{x} = \sum_{h=1}^H \mathbf{s}_h(f - (h - \frac{1}{2} - \frac{H}{2})f_{step})$. Then, the total bandwidth after synthesis is $B_p = HB_H$.

B. Dictionary Learning and Statistical Modeling of the Full-Band Frequency Data in Each Aspect-Frame

As discussed in the literatures [17], [18], HRRs from complex targets yield target signatures that are strongly dependent on the target-sensor orientation, referred to as target-aspect sensitivity [17]–[21]. To deal with the target-aspect sensitivity, the multi-aspect HRR dataset from each subset are collected from a target-aspect sector roughly without the scatterers' motion through range cells (MTRC) [17]–[20]. Thus each subset is defined to be an aspect-frame from the target [17]–[20].

The CS theory exploits the fact that $\mathbf{x}$ should be sparsely represented in an appropriate dictionary. Most of the CS literature assumes the dictionary is fixed and given. According to the scattering center model, an HRR represents the distributions of target scattering centers in the radial range dimension. Strong scattering centers usually take up only a fraction of whole range bins, while signals from weak scattering centers contribute little to the HRR formation [22]. Therefore, the echoed signal can be approximated by strong scattering centers, and that from weak ones is regarded as noise. In this sense, the HRR signal is sparse in the range domain, and the common Fourier dictionary is adopted for the CS-based HRR [4], [9]. However, recent research has demonstrated the significant utility of learning a dictionary matched to the signals of interest [23]–[25]. Specially, dictionary learning can also be cast as a factor analysis (FA) problem, with the factor loading corresponding to the dictionary elements (atoms) [23]. For the HRRs in one aspect-frame, since the scatterers are localized immovably in each range cell, there are strong correlations (some similarities) across the HRRs [18]. Therefore, the HRRs in one aspect-frame can share atoms in the learned dictionary. In other words, they can be sparsely represented in this dictionary.

For the traditional FA model [19], the number of factors, K , is preset and fixed, however, this may lead to overfitting if set too high or underfitting if set too low. Recent papers [23], [26], [27] propose to use the Beta-Bernoulli prior for inferring the number of factors in the FA model with application to high-dimensional data analysis, compressive sensing and image processing. However, these papers only consider the real-data applications. Actually, the statistical modeling of the complex data is highly significant for the radar signal. Therefore, we adopt a similar nonparametric Bayesian model CBPFA to accomplish the dictionary learning and statistical modeling of the complex full-band frequency data in each aspect-frame.

The generative CBPFA model of the n th full-band frequency sample in the m th frame of the target v (here $n \in \{1, 2, \dots, N\}$ with N denoting the number of samples in each aspect-frame, $m \in \{1, 2, \dots, M_v\}$ with M_v denoting the number of the aspect-frames of the v th target, and $v \in \{1, 2, \dots, V\}$ with V denoting the number of the targets) is given by:

$$\mathbf{x}_n^{(m,v)} = \mathbf{A}^{(m,v)} \mathbf{w}_n^{(m,v)} + \boldsymbol{\varepsilon}_n^{(m,v)}, \quad \mathbf{w}_n^{(m,v)} = \hat{\mathbf{w}}_n^{(m,v)} \odot \mathbf{z}_n^{(m,v)} \quad (8)$$

where $\mathbf{x}_n^{(m,v)} = [x_{n1}^{(m,v)}, x_{n2}^{(m,v)}, \dots, x_{nL}^{(m,v)}]^T$ with L denoting the number of the frequency components, $\mathbf{A}^{(m,v)} = [\mathbf{A}_1^{(m,v)}, \mathbf{A}_2^{(m,v)}, \dots, \mathbf{A}_K^{(m,v)}] \in C^{L \times K}$ represents the shared factor loading matrix or dictionary of the m th

frame of the target v with $\mathbf{A}_k^{(m,v)}$ denoting the k th column (element) of $\mathbf{A}^{(m,v)}$, K is an integer chosen to be the number of anticipated factors, $k \in \{1, 2, \dots, K\}$, $\boldsymbol{\varepsilon}_n^{(m,v)} = [\varepsilon_{n1}^{(m,v)}, \varepsilon_{n2}^{(m,v)}, \dots, \varepsilon_{nL}^{(m,v)}]^T \in C^{L \times 1}$ represents the noise variable, $\mathbf{w}_n^{(m,v)} = [w_{n1}^{(m,v)}, w_{n2}^{(m,v)}, \dots, w_{nK}^{(m,v)}]^T \in C^{K \times 1}$ denotes the latent factor variable or sparse weights, which has two components: complex valued weights $\hat{\mathbf{w}}_n^{(m,v)} = [\hat{w}_{n1}^{(m,v)}, \hat{w}_{n2}^{(m,v)}, \dots, \hat{w}_{nK}^{(m,v)}]^T$ and binary valued assignments $\mathbf{z}_n^{(m,v)} = [z_{n1}^{(m,v)}, z_{n2}^{(m,v)}, \dots, z_{nK}^{(m,v)}]^T$, the symbol $\odot$ represents the Hadamard product, or element-wise multiplication of two vectors. $\mathbf{z}_n^{(m,v)} \in \{0, 1\}$ is a binary vector that encodes which dictionary elements are activated for the corresponding observation and realizes the selection of the number of factors automatically.

In (8), $\boldsymbol{\varepsilon}_n^{(m,v)}$ is the independent noise, which is assumed to be the zero-mean complex Gaussian distributed with covariance $\tau^{(m,v)-1} \mathbf{I}_L$:

$$\begin{aligned} \boldsymbol{\varepsilon}_n^{(m,v)} &\sim \text{CN} \left(0, \tau^{(m,v)-1} \mathbf{I}_L \right) \\ &= \frac{1}{\pi^L |\tau^{(m,v)-1} \mathbf{I}_L|} \exp \left(-\boldsymbol{\varepsilon}_n^{(m,v)H} \left(\tau^{(m,v)-1} \mathbf{I}_L \right)^{-1} \boldsymbol{\varepsilon}_n^{(m,v)} \right) \end{aligned} \quad (9)$$

where $\mathbf{I}_L$ is the $L \times L$ dimensional identity matrix, and $\text{CN}(\cdot)$ specifies the complex Gaussian probability density distribution.

We first define a zero-mean complex Gaussian prior on $\hat{\mathbf{w}}_n^{(m,v)}$ [23], [26], [27], that is:

$$\hat{\mathbf{w}}_n^{(m,v)} \sim \text{CN} \left(0, \alpha^{(m,v)-1} \mathbf{I}_K \right) \quad (10)$$

To complete the specification of the hierarchical prior, we place Gamma priors on the precisions of the weights and observation noise ($\alpha^{(m,v)}$ and $\tau^{(m,v)}$) [23], [26], [27]:

$$\begin{aligned} \alpha^{(m,v)} &\sim \text{Gamma}(a_0, b_0) = \Gamma(a_0)^{-1} b_0^{a_0} \alpha^{(m,v)a_0-1} e^{-b_0 \alpha^{(m,v)}} \\ \tau^{(m,v)} &\sim \text{Gamma}(c_0, d_0) \end{aligned} \quad (11)$$

with $\Gamma(a) = \int_0^\infty t^{a-1} e^{-t} dt$ denoting the gamma function.

To model each dictionary element, we use a zero-mean complex Gaussian distribution:

$$\mathbf{A}^{(m,v)} \sim \prod_{k=1}^K \text{CN} \left(0, \frac{1}{L} \mathbf{I}_L \right) \quad (12)$$

For the binary vector $\mathbf{z}_n^{(m,v)}$, we use a finite Beta-Bernoulli approximation [23], [26], [27]. The finite model truncates the number of dictionary elements or factors to K and is given by:

$$\begin{aligned} \mathbf{z}_n^{(m,v)} &\sim \prod_{k=1}^K \text{Bernoulli} \left(u_k^{(m,v)} \right) \\ &= \prod_{k=1}^K u_k^{(m,v)z_{nk}^{(m,v)}} \left(1 - u_k^{(m,v)} \right)^{1-z_{nk}^{(m,v)}} \\ \mathbf{u}^{(m,v)} &\sim \prod_{k=1}^K \text{Beta} \left(\frac{e_0}{K}, \frac{f_0(K-1)}{K} \right) \end{aligned} \quad (13)$$

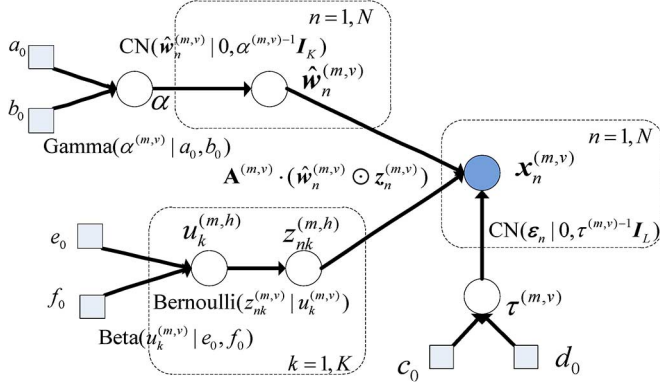


Fig. 1. Graphical model of the CBPFA for the full-band frequency data from each aspect-frame.

where e_0 and f_0 are scalars. As K tends to infinity, the finite Beta-Bernoulli approximation approaches the Beta-Bernoulli process. If the truncation is large enough, data analyzed with this prior will exhibit fewer than K components [28].

Through the aforementioned introduction of the model and the detailed parameters, we now have a complete probabilistic specification of this model, with a graphical model representation illustrated in Fig. 1.

C. Variational Bayesian Inference

The CBPFA model may be readily implemented via Markov Chain Monte Carlo (MCMC) sampling [29], [30]. However, we employ variational Bayesian (VB) [30]–[32] inference in this paper, as a compromise between accuracy and efficiency.

Bayesian inference seeks to estimate the posterior distributions of the latent variables Ψ in the model, given the observed data $\mathbf{D}$ and hyper-parameters θ . We seek a variational distribution $q(\Psi) = \prod_f q(\Psi_f)$ to approximate the true posterior distribution of the latent variables $p(\Psi|\mathbf{D}, \theta)$. The expression is:

$$\ln p(\mathbf{D}|\theta) = L(q(\Psi)) + \text{KL}(q(\Psi)|p(\Psi|\mathbf{D}, \theta)) \quad (14)$$

where $\text{KL}(q(\Psi)|p(\Psi|\mathbf{D}, \theta))$ is Kullback-Leibler divergence between the variational approximation $q(\Psi)$ and the true joint posterior $p(\Psi|\mathbf{D}, \theta)$. Since $\text{KL}(q(\Psi)|p(\Psi|\mathbf{D}, \theta)) \geq 0$, it follows that $L(q(\Psi))$ is a rigorous lower bound on $\ln p(\mathbf{D}|\theta)$. Accordingly, the optimal parameterization of $q(\Psi_f)$ is obtained by minimizing $\text{KL}(q(\Psi)|p(\Psi|\mathbf{D}, \theta))$, i.e., maximizing the similarity between the variational distribution and the true posterior.

In our CBPFA model, $\mathbf{D}$ is $\{\mathbf{x}_n^{(m,v)}\}_{n=1}^N$, Ψ is $\{\mathbf{A}^{(m,v)}, \hat{\mathbf{w}}_n^{(m,v)}, \mathbf{z}_n^{(m,v)}, u_k^{(m,v)}, \alpha^{(m,v)}, \tau^{(m,v)}\}_{n=1}^N$, and θ is $\{a_0, b_0, c_0, d_0, e_0, f_0\}$. A general method for performing the variational inference for a conjugate-exponential Bayesian network is as follows [33]: For a given node in a graphic model, write out the posterior as though everything were known, take the logarithm, the expectation with respect to all unknown parameters and exponentiate the result. The algorithm iteratively updates $\{q(\Psi_f)\}$ so that $L(q(\Psi))$

approaches to $\ln p(\mathbf{D}|\theta)$ until convergence. The VB inference for the model is summarized in Algorithm 1.

Algorithm 1: VB-based Inference for CBPFA

Input: the full-band frequency samples in an aspect-frame $\mathbf{D} = \{\mathbf{x}_n^{(m,v)}\}_{n=1}^N$; the initial hyper-parameters θ_0 : $\alpha_0 = b_0 = c_0 = d_0 = 10^{-6}$, $e_0 = f_0 = 1$, and $K = 100$; a small positive constant $\zeta = 10^{-6}$.

Output: the positive distributions of the latent variables Ψ : $q(\hat{\mathbf{w}}_n^{(m,v)})$, $q(\alpha^{(m,v)})$, $q(A_{kl}^{(m,v)})$, $q(\tau^{(m,v)})$, $q(z_{nk}^{(m,v)})$, and $q(u_k^{(m,v)})$; the lower bound L .

Initialization: $\theta \leftarrow \theta_0$; $iter = 0$; $change = 1$;

$L(1) = \int q(\Psi) \ln(p(\mathbf{D}, \Psi)/q(\Psi)) d\Psi$

While $change > \zeta$

$iter = iter + 1$

$q(\hat{\mathbf{w}}_n^{(m,v)}) = \text{CN}(\mu_n^{(m,v)}, \Sigma_n^{(m,v)})$,

$\Sigma_n^{(m,v)} = (\langle \alpha^{(m,v)} \rangle \mathbf{I}_K + \langle \tau^{(m,v)} \rangle \langle \mathbf{z}_n^{(m,v)} \mathbf{z}_n^{(m,v)T} \rangle \odot \langle \mathbf{A}^{(m,v)H} \mathbf{A}^{(m,v)} \rangle)^{-1}$
 $\mu_n^{(m,v)} = \langle \tau^{(m,v)} \rangle \Sigma_n^{(m,v)} \text{diag}(\langle \mathbf{z}_n^{(m,v)} \rangle) \langle \mathbf{A}^{(m,v)H} \rangle \mathbf{x}_n^{(m,v)}$ (15)
 $q(\alpha^{(m,v)}) = \text{Gamma}(\tilde{a}, \tilde{b})$

$\tilde{a} = a_0 + KN$, $\tilde{b} = b_0 + \sum_{n=1}^N \sum_{k=1}^K \langle \hat{w}_{nk}^{(m,v)2} \rangle$ (16)

$q(A_{kl}^{(m,v)}) = \text{CN}(\varsigma_{kl}^{(m,v)}, B_{kl}^{(m,v)})$

$B_{kl}^{(m,v)} = (L + \sum_{n=1}^N \langle \tau^{(m,v)} \rangle \langle z_{nk}^{(m,v)} \rangle \langle \hat{w}_{nk}^{(m,v)2} \rangle)^{-1}$

$\varsigma_k^{(m,v)} = B_k^{(m,v)} (\sum_{n=1}^N \langle \tau^{(m,v)} \rangle \langle z_{nk}^{(m,v)} \rangle \langle \hat{w}_{nk}^{(m,v)H} \rangle (\mathbf{x}_n^{(m,v)} - \sum_{h \neq k} z_{nh}^{(m,v)} \langle \hat{w}_{nh}^{(m,v)} \rangle \mathbf{A}_h^{(m,v)}))$ (17)

$q(\tau^{(m,v)}) = \text{Gamma}(\tilde{c}, \tilde{d})$, $\tilde{c} = c_0 + NL$

$\tilde{d} = d_0 + \sum_{n=1}^N (\mathbf{x}_n^{(m,v)H} \mathbf{x}_n^{(m,v)} - 2 \mathbf{x}_n^{(m,v)H} \langle \mathbf{A}^{(m,v)} \rangle \times \text{diag}(\mathbf{z}_n^{(m,v)}) \langle \hat{\mathbf{w}}_n^{(m,v)} \rangle + \text{sum}(\text{diag}((\langle \hat{\mathbf{w}}_n^{(m,v)} \hat{\mathbf{w}}_n^{(m,v)H} \rangle \odot \langle \mathbf{z}_n^{(m,v)} \mathbf{z}_n^{(m,v)T} \rangle) \langle \mathbf{A}^{(m,v)H} \mathbf{A}^{(m,v)} \rangle)))$ (18)

$q(z_{nk}^{(m,v)}) = \text{Bernoulli}(\frac{p(z_{nk}^{(m,v)} = 1)}{p(z_{nk}^{(m,v)} = 0) + p(z_{nk}^{(m,v)} = 1)})$ (19)

$q(u_k^{(m,v)}) = \text{Beta}(\frac{\tilde{e}_k}{K}, \frac{\tilde{f}_k(K-1)}{K})$

$\tilde{e}_k = e_0 + K \sum_{n=1}^N \langle z_{nk}^{(m,v)} \rangle$,

$\tilde{f}_k = f_0 + \frac{K}{K-1} (N - \sum_{n=1}^N \langle z_{nk}^{(m,v)} \rangle)$ (20)

$L(iter + 1) = \int q(\Psi) \ln \frac{p(\mathbf{D}, \Psi)}{q(\Psi)} d\Psi$

$change = \frac{(L(iter + 1) - L(iter))}{L(iter)}$

End

In Algorithm 1, $\langle \cdot \rangle$ means the posterior expectation for the latent variable over the corresponding distribution on it. In (19), the posterior of $z_{nk}^{(m,v)}$ is $q(z_{nk}^{(m,v)}) = \text{Bernoulli}(p(z_{nk}^{(m,v)} = 1)/(p(z_{nk}^{(m,v)} = 0) + p(z_{nk}^{(m,v)} = 1)))$ with the probability that $z_{nk}^{(m,v)} = 1$ proportional to

$$\begin{aligned} p(z_{nk}^{(m,v)} = 1) &\propto \exp(\langle \ln(u_k^{(m,v)}) \rangle) \\ &\times \exp(\langle \tau^{(m,v)} \rangle \langle 2 \langle \hat{w}_k^{(m,v)H} \rangle \langle \mathbf{A}_k^{(m,v)H} \rangle \\ &\times \left(\mathbf{x}_n^{(m,v)} - \sum_{h \neq k} z_{nh}^{(m,v)} \langle \hat{w}_{nh}^{(m,v)} \rangle \langle \mathbf{A}_h^{(m,v)} \rangle \right) \\ &- \langle \hat{w}_{nk}^{(m,v)2} \rangle \langle \mathbf{A}_k^{(m,v)H} \mathbf{A}_k^{(m,v)} \rangle) \end{aligned}$$

and the probability that $z_{nk}^{(m,v)} = 0$ proportional to $p(z_{nk}^{(m,v)} = 0) \propto \exp(\langle \ln(1 - u_k^{(m,v)}) \rangle)$, in which $\langle \ln(u_k^{(m,v)}) \rangle$ and $\langle \ln(1 - u_k^{(m,v)}) \rangle$ are used, they can be calculated according to the update of $u_k^{(m,v)}$ in (20) at the bottom of the next page:

$$\begin{aligned} \langle \ln(u_k^{(m,v)}) \rangle &= \psi\left(\frac{\tilde{e}_k}{K}\right) - \psi\left(\frac{\tilde{e}_k + \tilde{f}_k(K-1)}{K}\right) \\ \langle \ln(1 - u_k^{(m,v)}) \rangle &= \psi\left(\frac{\tilde{f}_k(K-1)}{K}\right) \\ &- \psi\left(\frac{\tilde{e}_k + \tilde{f}_k(K-1)}{K}\right) \end{aligned} \quad (21)$$

where $\psi(\cdot)$ represents the digamma function.

At convergence of the parameter estimation, we can obtain the posterior expectations $\langle \mathbf{A}^{(m,v)} \rangle$, $\langle \hat{w}_n^{(m,v)} \rangle$, $\langle \mathbf{z}_n^{(m,v)} \rangle$, and $\langle \tau^{(m,v)} \rangle$. Specially, $\mathbf{z}_n^{(m,v)}$ is a binary vector, if $q(z_{nk}^{(m,v)} = 1) = \langle z_{nk}^{(m,v)} \rangle \geq 0.5$, then $z_{nk}^{(m,v)} = 1$, otherwise, $z_{nk}^{(m,v)} = 0$. According to the nonzero elements of $\mathbf{z}_n^{(m,v)}$, the corresponding elements of $\mathbf{A}^{(m,v)}$ and $\hat{w}_n^{(m,v)}$ can be selected, and the number of factors is inferred automatically. In addition, the update equations of $\tilde{d}$ and $p(z_{nk}^{(m,v)} = 1)$ are invariant to each sample's initial-phase due to the conjugate product calculation. Thus the initial-phase sensitivity problem can be avoided in the model learning procedure.

D. Probability Density Estimation for the Full-Band Frequency Data in Each Aspect-Frame

After the signal model is inferred for the full-band frequency samples from a given aspect-frame, the pdf of this aspect-frame can be estimated as follows:

$$\begin{aligned} p(\mathbf{x}|m, v) &= \int \text{CN}(\mathbf{x}; \langle \mathbf{A}^{(m,v)} \rangle \mathbf{w}^{(m,v)}, \langle \tau^{(m,v)} \rangle^{-1} \mathbf{I}_L) \\ &\times \text{CN}(\mathbf{w}^{(m,v)}; \boldsymbol{\mu}^{(m,v)}, \boldsymbol{\Sigma}^{(m,v)}) d\mathbf{w} \end{aligned} \quad (22)$$

where $\langle \mathbf{A}^{(m,v)} \rangle$ and $\langle \tau^{(m,v)} \rangle$ have been obtained in part C of Section II. However, in (8), different samples have different sparse weights $\mathbf{w}_n^{(m,v)} = \hat{\mathbf{w}}_n^{(m,v)} \odot \mathbf{z}_n^{(m,v)}$ in one aspect-frame. Therefore, to estimate the pdf of this aspect-frame, we should first calculate the statistical weights

$\mathbf{w}^{(m,v)}$ of this aspect-frame. Since the posterior distributions $q(\hat{\mathbf{w}}_n^{(m,v)}) = \text{CN}(\boldsymbol{\mu}_n^{(m,v)}, \boldsymbol{\Sigma}_n^{(m,v)})$ and $\mathbf{z}_n^{(m,v)}$ have been obtained, we can use the following posterior mean and covariance to represent the statistical property of $\mathbf{w}^{(m,v)}$ from Gibbs sampling:

$$\begin{aligned} \boldsymbol{\mu}^{(m,v)} &= \frac{1}{N} \sum_{n=1}^N (\langle \hat{\mathbf{w}}_n^{(m,v)} \rangle \odot \mathbf{z}_n^{(m,v)}) \\ \boldsymbol{\Sigma}^{(m,v)} &= \frac{1}{N} \sum_{n=1}^N \left((\langle \hat{\mathbf{w}}_n^{(m,v)} \rangle \odot \mathbf{z}_n^{(m,v)} - \boldsymbol{\mu}^{(m,v)}) \right. \\ &\quad \times \left. \left((\langle \hat{\mathbf{w}}_n^{(m,v)} \rangle \odot \mathbf{z}_n^{(m,v)} - \boldsymbol{\mu}^{(m,v)})^H \right) \right) \end{aligned} \quad (23)$$

It is reasonable to assume that $\mathbf{w}^{(m,v)}$ follows a complex Gaussian distribution with zero mean [34], and which can also be verified by the experimental results. Therefore, to simplify the model, the distribution of $\mathbf{w}^{(m,v)}$ can be written as:

$$\begin{aligned} q(\mathbf{w}^{(m,v)}) &= \text{CN}(\mathbf{0}, \boldsymbol{\Sigma}^{(m,v)}) \\ \boldsymbol{\Sigma}^{(m,v)} &= \frac{1}{N} \sum_{n=1}^N \left((\langle \hat{\mathbf{w}}_n^{(m,v)} \rangle \odot \mathbf{z}_n^{(m,v)}) \right. \\ &\quad \times \left. \left((\langle \hat{\mathbf{w}}_n^{(m,v)} \rangle \odot \mathbf{z}_n^{(m,v)})^H \right) \right) \end{aligned} \quad (24)$$

Due to the conjugacy, we can obtain the pdf of this given aspect-frame in (22):

$$\begin{aligned} p(\mathbf{x}|m, v) &= \text{CN}(\mathbf{0}, \boldsymbol{\Lambda}^{(m,v)}) \\ \boldsymbol{\Lambda}^{(m,v)} &= \langle \mathbf{A}^{(m,v)} \rangle \boldsymbol{\Sigma}^{(m,v)} \langle \mathbf{A}^{(m,v)} \rangle^H \\ &\quad + \langle \tau^{(m,v)} \rangle^{-1} \mathbf{I}_L \end{aligned} \quad (25)$$

In this paper, for the auxiliary/training data, we assume it contains V targets and each target includes M_v aspect-frames. Therefore, there are $T = \sum_{v=1}^V M_v$ training aspect-frames in all. In addition, for convenience, we use t to replace (m, v) , then $t = (m, v)$, and $t \in \{1, 2, \dots, T\}$.

III. STATISTICAL COMPRESSIVE SENSING OF THE INCOMPLETE FREQUENCY DATA BASED ON TRANSFER LEARNING

In the reconstruction stage, for the obtained incomplete frequency data $\mathbf{y}^*$, which is the summation of the subset of $\{\mathbf{s}_h(f - (h - \frac{1}{2} - \frac{H}{2})f_{step})\}_{h=1}^H$. Assuming its corresponding full-band frequency data is $\mathbf{x}^*$. Then, their relationship can be expressed as $\mathbf{y}^* = \boldsymbol{\Phi} \mathbf{x}^* + \boldsymbol{\xi}^*$, where $\boldsymbol{\Phi}$ is the extracting matrix, whose elements are 1 or 0, and $\boldsymbol{\xi}^*$ is the noise. Thus, our goal is to describe a general-purpose algorithm for recovering $\mathbf{x}^*$ based on the incomplete frequency sample $\mathbf{y}^*$ and the learned pdfs $\{p(\mathbf{x}|m, v)\}_{t=(m,v)=1}^T$ of the training data in (25).

We assume the noise $\boldsymbol{\xi}^*$ is drawn from a zero-mean complex Gaussian with covariance $\vartheta^{-1} \mathbf{I}_J$. Thus, $p(\mathbf{y}^*|\mathbf{x}^*) = \text{CN}(\mathbf{y}^*, \boldsymbol{\Phi} \mathbf{x}^*, \vartheta^{-1} \mathbf{I}_J)$.

For the HRR data, the characteristics of different aspect-frame vary greatly, in other words, the relatedness between each training aspect-frame and $\mathbf{y}^*$ is changed with the variation of the frame label $t = (m, v)$. Therefore, we utilize the frame

condition distribution of $\mathbf{y}^*$ to represent the relatedness between each training aspect-frame and $\mathbf{y}^*$:

$$\begin{aligned} p(\mathbf{y}^*|m, v) &= \int p(\mathbf{y}^*|\mathbf{x}^*)p(\mathbf{x}^*|m, v)d\mathbf{x}^* \\ &= \text{CN}(\mathbf{y}^*; 0, \vartheta^{-1}\mathbf{I}_J + \Phi\Lambda^{(m,v)}\Phi^H) \end{aligned} \quad (26)$$

Then, the condition distribution for $\mathbf{x}^*$ given $\mathbf{y}^*$ with each aspect-frame can be analytically calculated [35]:

$$\begin{aligned} p(\mathbf{x}^*|\mathbf{y}^*, m, v) &= \frac{p(\mathbf{x}^*|m, v)p(\mathbf{y}^*|\mathbf{x}^*)}{\int p(\mathbf{x}^*|m, v)p(\mathbf{y}^*|\mathbf{x}^*)d\mathbf{x}^*} \\ &= \frac{\text{CN}(\mathbf{x}^*; 0, \Lambda^{(m,v)}) \times \text{CN}(\mathbf{y}^*; \Phi\mathbf{x}^*, \vartheta^{-1}\mathbf{I}_J)}{\int \text{CN}(\mathbf{x}^*; 0, \Lambda^{(m,v)}) \times \text{CN}(\mathbf{y}^*; \Phi\mathbf{x}^*, \vartheta^{-1}\mathbf{I}_J)d\mathbf{x}^*} \\ &= \text{CN}(\mathbf{x}^*; \tilde{\chi}^{(m,v)}, \tilde{\Lambda}^{(m,v)}) \end{aligned} \quad (27)$$

with

$$\begin{aligned} \tilde{\Lambda}^{(m,v)} &= (\vartheta\Phi^T\Phi + \Lambda^{(m,v)} - 1)^{-1} \\ &= \Lambda^{(m,v)} - \Lambda^{(m,v)}\Phi^T(\vartheta^{-1}\mathbf{I}_J + \Phi\Lambda^{(m,v)}\Phi^T)^{-1}\Phi\Lambda^{(m,v)} \\ \tilde{\chi}^{(m,v)} &= \Lambda^{(m,v)}\Phi^T(\vartheta^{-1}\mathbf{I}_J + \Phi\Lambda^{(m,v)}\Phi^T)^{-1}\mathbf{y}^* \end{aligned} \quad (28)$$

With the varying degree of relatedness between the incomplete frequency data $\mathbf{y}^*$ and each training aspect-frame, we can utilize the weighting of $\{p(\mathbf{x}^*|\mathbf{y}^*, m, v)\}_{t=(m,v)=1}^T$ to represent the condition distribution of $\mathbf{x}^*$ given $\mathbf{y}^*$:

$$\begin{aligned} p(\mathbf{x}^*|\mathbf{y}^*) &= \sum_{t=(m,v)=1}^T \pi_t p(\mathbf{x}^*|\mathbf{y}^*, m, v) \\ &= \sum_{t=(m,v)=1}^T \pi_t \text{CN}(\mathbf{x}^*; \tilde{\chi}^{(m,v)}, \tilde{\Lambda}^{(m,v)}) \end{aligned} \quad (29)$$

where $\{\pi_t\}_{t=1}^T$ are the weights, and they can be defined by the relatedness between each training aspect-frame and $\mathbf{y}^*$ given in (26). In addition, the weights $\{\pi_t\}_{t=1}^T$ must satisfy $0 \leq \pi_t \leq 1$ together with $\sum_{t=1}^T \pi_t = 1$ in order to be valid probabilities. Therefore, we utilize the normalized value of $p(\mathbf{y}^*|m, v)$ to represent π_t :

$$\pi_t = \frac{p(\mathbf{y}^*|m, v)}{\sum_{t=(m,v)=1}^T p(\mathbf{y}^*|m, v)} \quad (30)$$

Consequently, π_t provides the flexibility of no sharing when the $t = (m, v)$ th training aspect-frame and $\mathbf{y}^*$ are unrelated and partial/total sharing when they are related.

In the experiments presented below, we mainly consider the case that ϑ^{-1} tends to zero, i.e., assuming the noise-free measurements. In addition, since $p(\mathbf{x}^*|m, v)$ can be learned offline, we have closed-form expressions for all the parameters in the posterior distribution $p(\mathbf{x}^*|\mathbf{y}^*, m, v)$. Moreover, we formulate and solve the full distribution $p(\mathbf{x}^*|\mathbf{y}^*)$ within a completely

TABLE I
PARAMETER OF PLANES AND RADAR IN THE ISAR EXPERIMENT

radar parameters	center frequency		5520 MHz
	bandwidth		400 MHz
planes	length (m)	width (m)	height (m)
Yak-42	36.38	34.88	9.83
Cessna Citation S/II	14.40	15.90	4.57
An-26	23.80	29.20	9.83

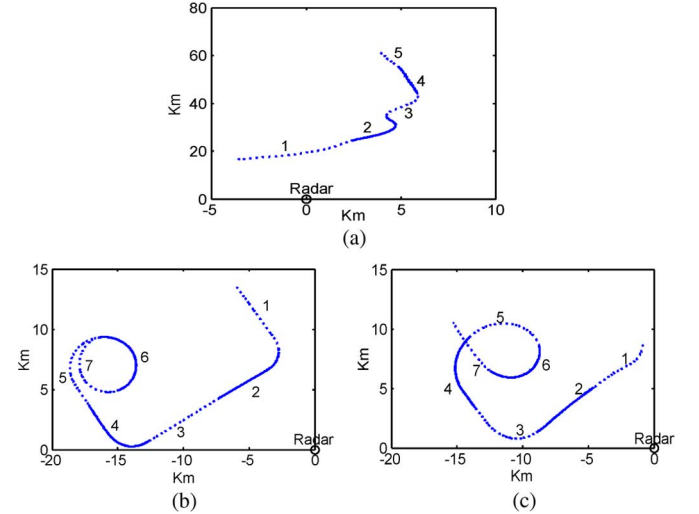


Fig. 2. Projections of target trajectories onto the ground plane. (a) Yak-42; (b) Cessna Citation S/II; (c) An-26.

Bayesian paradigm rather than simply providing a point estimate for $\mathbf{x}^*$. In the actual operation, we can take the mean value $\sum_{t=(m,v)=1}^T \pi_t \tilde{\chi}^{(m,v)}$ as the reconstruction of $\mathbf{x}^*$.

IV. EXPERIMENTAL RESULTS

Since the proposed algorithm is applied to the frequency data after the synthesis of the stepped-frequency signal, without considering the synthetic procedure. Therefore, in this section, we directly examine the performance of the proposed algorithm on the measured HRR data from three real airplanes, which are also used in [17], [18], [20], [33], [36]. The parameters of the airplane targets and measurement radar are shown in Table I. The projections of the target trajectories onto the ground plane are shown in Fig. 2, from which the aspect angle of an airplane can be estimated according to its relative position to radar. As shown in Fig. 2, the measured data are segmented, where each segment contains 26000 HRR samples and each sample is a 256-length vector, i.e., $L = 256$. In our reconstruction experiments, data from the 2nd segment and a part of the 5th segment of Yak-42, the 6th and the 7th segments of Cessna Citation S/II, the 5th and the 6th segments of An-26 are taken as the training samples, and other data segments are taken as test samples to evaluate the proposed reconstruction algorithm.

A. Training Results Analysis of the Full-Band Frequency Data

It is a prerequisite for HRR to deal with the target-aspect, time-shift, and amplitude-scale sensitivity problems [17], [18].

We have discussed the target-aspect problem in part B of Section II. Same as the previous work [17]–[19], [36], we use

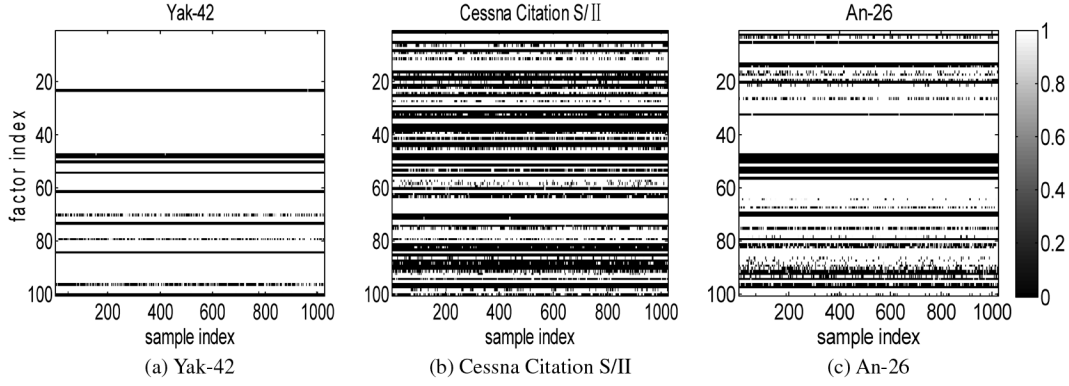


Fig. 3. (a) Factor usage of each sample in one aspect-frame from Yak-42. (b) Factor usage of each sample in one aspect-frame from Cessna Citation S/II. (c) Factor usage of each sample in one aspect-frame from An-26.

the aspect-frame partition method to deal with the target-aspect sensitivity of the training dataset. Therefore, in training stage, there are 35, 50, and 50 aspect-frames for Yak-42, Cessna Citation S/II, and An-26, respectively, of which each aspect-frame totally has 1024 HRR samples, i.e. $V = 3$, $M_1 = 35$, $M_2 = M_3 = 50$, $N = 1024$, $T = 135$.

About the time-shift and amplitude-scale sensitivity problems. Since an HRR sample is only a part of the received radar echo extracted by a range window, in which target signal is included, the position of the target signal in an HRR sample can vary with the measurement. This is the time-shift sensitivity. To deal with the time-shift sensitivity, time-shift compensation techniques in ISAR imaging, such as the slide correlation processing or its modifications [22] can be used. Since the intensity of an HRR sample is a function of radar transmitting power, target distance, radar antenna gain, radar receiver gain, radar system losses and so on, HRR samples measured by different radars or under different conditions will have different amplitude-scales. To deal with the amplitude-scale sensitivity, the L_2 amplitude normalization can be used as a preprocessing procedure [18], [19].

The proposed CBPFA model can infer the proper number of factors for each aspect-frame of the training data, in which $\mathbf{z}_n^{(m,v)}$ is a binary vector used to achieve the sparseness. Fig. 3 shows three cases of $\{\langle \mathbf{z}_n^{(m,v)} \rangle\}_{n=1}^{1024}$ for Yak-42, Cessna Citation S/II, and An-26, respectively. From Fig. 3, we can see that the number of nonzero elements of $\langle \mathbf{z}_n^{(m,v)} \rangle$ is obviously smaller than the number of anticipated factors $K = 100$ for all of the three targets, which justifies the automatic inference of the number of the factors by using the BP prior. Fig. 4 presents the number of factors of each aspect-frame from the three targets. In Fig. 4, the inferred number of factors varies across the frame index, which shows the utility of using such nonparametric model selection method. By contrast, if we fix the same number of factors for all frames, which may lead to overfitting or underfitting for some frames.

B. CS Reconstruction Performance of Non-Uniform Piecewise Extraction

After accomplishing the training procedure, now we present an experiment to demonstrate the reconstruction performance

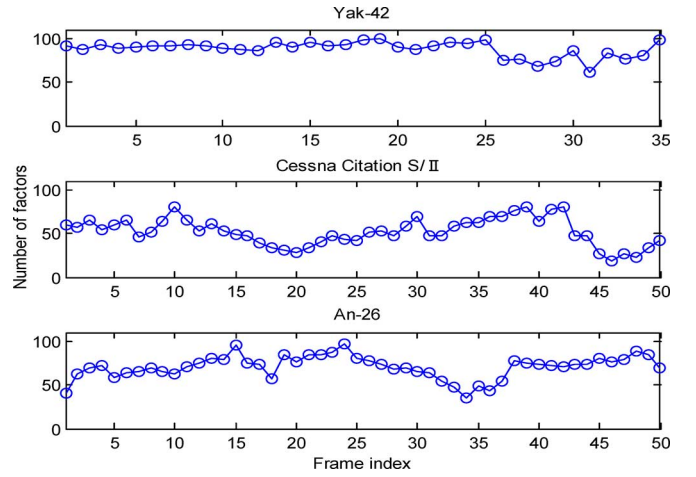


Fig. 4. Number of factors per aspect-frame estimated via our proposed method.

of the proposed algorithm. Here we use 3000 HRR test samples for compressive sensing and reconstruction. Each target has 1000 samples, which averagely come from all of the testing aspect-frames to guarantee the universality of the proposed method. We utilize the mean relative reconstruction error, $\frac{\sum_{i=1}^I \|\langle \mathbf{x}_i^* | \mathbf{y}_i^* \rangle - \mathbf{x}_i^*\|_2}{\sum_{i=1}^I \|\mathbf{x}_i^*\|_2}$, to quantify the reconstruction performance, where I represents the number of the test samples, $\mathbf{y}_i^* \in C^{J \times 1}$ represents the i th observed incomplete HRR frequency data with $i \in \{1, 2, \dots, I\}$, $\mathbf{x}_i^* \in C^{L \times 1}$ denotes the corresponding true full-band HRR frequency data, $\mathbf{y}_i^* = \Phi \mathbf{x}_i^* + \xi_i^*$, and $\langle \mathbf{x}_i^* | \mathbf{y}_i^* \rangle$ represents the conditional expectation of the i th reconstructed full-band HRR frequency data given the measurement data $\mathbf{y}_i^*$.

In the Introduction, we have explained why set Φ as an extracting matrix. However, there is not a conclusion about how to extract to obtain the optimal reconstruction performance. Therefore, in this part, we use the most common non-uniform piecewise extraction, i.e., randomly extracting some pulses of the stepped-frequency chirp signal.

Firstly, all of the 135 training aspect-frames are taken as the training data, whose classes include those of the test samples. Now, we investigate how the number of measurements influences the reconstruction performance. We vary the number of

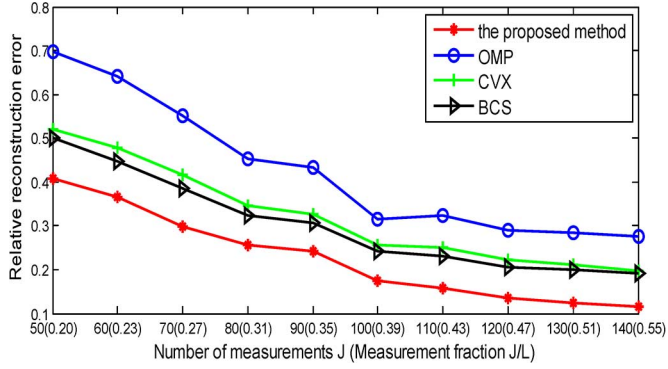


Fig. 5. The mean relative reconstruction errors of the 3000 test samples versus the numbers of measurements J (measurement fraction J/L) for non-uniform piecewise extraction. The comparative approaches include our proposed method, OMP, CVX, and BCS. For our method, all of 135 training frames are used as the training data.

measurements J from 50 to 140 (the interval is 10), i.e., the corresponding measurement fractions J/L ($L = 256$) are: 0.20, 0.23, 0.27, 0.31, 0.35, 0.39, 0.43, 0.47, 0.51, and 0.55. Specifically, we compare our method with three traditional CS recovery approaches: orthogonal matching pursuit (OMP) [10], convex optimization based on $L1$ norm minimization [37] (it is solved by CVX¹ toolbox), and Bayesian compressive sensing (BCS²) [12], of which BCS is implemented by the VB analysis. For these three methods, the dictionary $\mathbf{Q}$ is chosen as the Fourier dictionary according to the scattering property of the HRR data [4], [9], whose dimensionality is $E = 800$.

Fig. 5 gives the variations of the mean relative reconstruction errors of the 3000 test samples with the numbers of the measurements J (measurement fraction J/L) for the four methods. As seen in Fig. 5, our method outperforms OMP, CVX, and BCS, which shows the benefits of leveraging the transfer learning and the automatically Bayesian dictionary learning. Moreover, the reconstruction performance of BCS is better than those of OMP and CVX, which also verify the efficiency of the Bayesian method to some extent. In addition, with the increase of the number of measurements J , the relative reconstruction errors of the four methods tend to decrease, which demonstrates that the larger the number of the measurements is, the better the reconstruction performance will be to a certain extent.

The reconstruction results of two HRR examples with the four methods are given in Fig. 6. In order to verify the reconstruction accuracy of the amplitude and phase of the HRR frequency data simultaneously, we transform the reconstructed HRR frequency data to the HRR time domain through the inverse fast Fourier transform (IFFT). Here we only give two cases of the ten measurement fractions in Fig. 5: 0.27 and 0.47. In Fig. 6, the left column shows one reconstruction example when measurement fraction is 0.27, and the right column shows another reconstruction example when measurement fraction is 0.47, different methods for each row. From Fig. 6, we can see

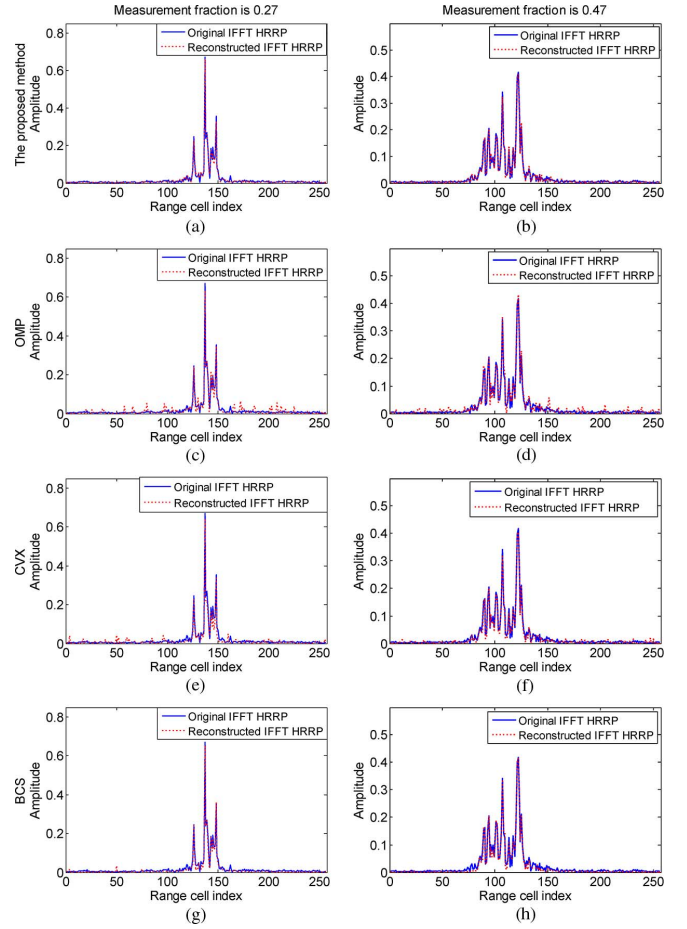


Fig. 6. Reconstruction examples of the HRR data for non-uniform piecewise extraction. The left column shows one reconstruction IFFT HRRP example when measurement fraction is 0.27 (different methods for each row), and the right column shows another reconstruction IFFT HRRP example when measurement fraction is 0.47. For our method, all of 135 training frames are used as the training data. The relative reconstruction errors of the examples are: (a) The proposed method: 0.1499, (b) The proposed method: 0.1070, (c) OMP: 0.2836, (d) OMP: 0.2366, (e) CVX: 0.2626, (f) CVX: 0.1557, (g) BCS: 0.2412, (h) BCS: 0.1666.

that no matter whether the measurement fraction is 0.27 or 0.47, our method outperforms the three competing methods with better reconstruction performance. Moreover, there are spurious peaks in the IFFT high range resolution profile (HRRP) reconstructed by OMP compared with our method, BCS, and CVX.

When the measurement fraction is 0.27, the normalized transfer weights $\{\pi_t\}_{t=1}^T$ of 3000 test samples in our method are depicted in Fig. 7. As seen in Fig. 7, the test samples in one category may be reconstructed by utilizing not only the training frames of the same category but also those of the other categories. For example, the 1000 test samples from Yak-42 are mainly reconstructed by the training frames from both Yak-42 and An-26. Therefore, the results verify the notion that the common/related information between one test sample and each training frame is employed to realize the reconstruction.

In Fig. 7, although the test samples can be reconstructed by the training frames of other categories, the training frames of the same category also play the major role. In order to verify

¹The toolbox of CVX is available from <http://stanford.edu/~boyd/cvx>.

²The software of VB-based BCS is coded by the first author of this paper,

TABLE II

THE MEAN RELATIVE RECONSTRUCTION ERRORS OF 1000 TEST SAMPLE FROM EACH CATEGORY BY OUR METHOD ($PM_{1,2}$, $PM_{1,3}$, $PM_{2,3}$, AND $PM_{1,2,3}$), OMP, CVX, AND BCS WHEN THE MEASUREMENT FRACTIONS ARE 0.27 AND 0.47

Methods	Measurement fraction is 0.27				Measurement fraction is 0.47			
	Yak-42 (1000)	Cessna Citation S/II (1000)	An-26 (1000)	Mean (3000)	Yak-42 (1000)	Cessna Citation S/II (1000)	An-26 (1000)	Mean (3000)
$PM_{1,2}$	\	\	0.3026	0.3521	\	\	0.1267	0.1644
$PM_{1,3}$	\	0.2296	\		\	0.1045	\	
$PM_{2,3}$	0.5241	\	\		0.2619	\	\	
$PM_{1,2,3}$	0.4617	0.1725	0.2570	0.2971	0.2201	0.0911	0.0925	0.1345
OMP	0.8212	0.3759	0.4579	0.5516	0.4662	0.2302	0.1696	0.2887
CVX	0.5892	0.3005	0.3559	0.4152	0.3501	0.1702	0.1471	0.2225
BCS	0.5627	0.2654	0.3307	0.3863	0.3119	0.1526	0.1548	0.2064

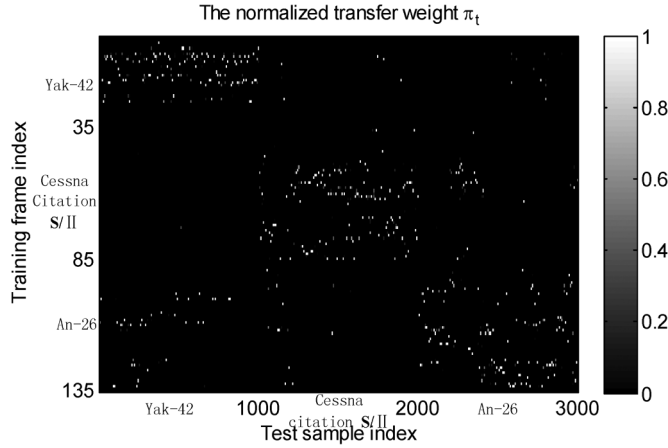


Fig. 7. When the measurement fraction is 0.27, the normalized transfer weights of 135 training frames for each test sample in our method for non-uniform piecewise extraction. On the horizontal ordinate, the first 1000 test samples belong to Yak-42, the subsequent 1000 test samples are Cessna Citation S/II, and the remainder belong to An-26. On the vertical ordinate, the first 35 frames are Yak-42, the subsequent 50 frames belong to Cessna Citation S/II, and the last 50 frames belong to An-26.

the universality of the proposed method, we present another experiment that the 1000 test samples of each category are only reconstructed by the training frames from the other two different categories, e.g., the test samples from Yak-42 are only reconstructed by 50 training frames from Cessna Citation S/II and 50 training frames from An-26.

Table II gives the mean relative reconstruction error of 1000 test samples from each category by our method ($PM_{1,2}$, $PM_{1,3}$, $PM_{2,3}$, and $PM_{1,2,3}$), OMP, CVX, and BCS when the measurement fractions are 0.27 and 0.47, in which $PM_{1,2}$ represents our proposed method when the training data is 35 frames from Yak-42 and 50 frames from Cessna Citation S/II, $PM_{1,3}$ denotes our method when the training data includes 35 frames from Yak-42 and 50 frames from An-26, $PM_{2,3}$ represents our method when the training data is 50 frames from Cessna Citation S/II and 50 frames from An-26, and $PM_{1,2,3}$ denotes our method when the training data includes all of the 135 training frames. From Table II, we can see that the reconstruction errors of $PM_{1,2,3}$ are lower than those of $PM_{1,2}$, $PM_{1,3}$, and $PM_{2,3}$ (the categories of the training data are different from that of the test samples), which indicates the sufficient categories and frames of the training data are of benefit to the reconstruction performance. Actually, in practical application, the training

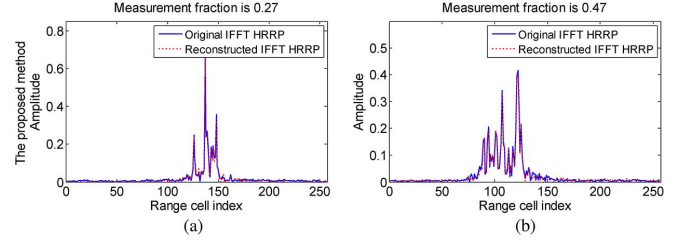


Fig. 8. When the categories of the test samples are different from those of the training samples for our method, the reconstruction results of two IFFT HRRP examples for non-uniform piecewise extraction. The relative reconstruction errors of the examples are: (a) 0.1688, (b) 0.1508.

database which is learned offline can be supplemented at any time, and it is reasonable and useful that the training database includes the categories of the test samples. In addition, even though the categories of the test samples are different from those of the training samples, our method still outperforms the other three comparative methods. Therefore, our method is universal to reconstruct an arbitrary test sample as long as it is “related” to the training samples, even they are with the different classes.

Following, when the categories of the test samples are different from those of the training samples for our method, the reconstruction results of two HRR samples same as those in Fig. 6 are given in Fig. 8. Comparing Fig. 8 with Fig. 6, we can see that the reconstruction results in Fig. 8 are still better than those of OMP, CVX, and BCS in Fig. 6, which demonstrates that our method is more appealing to realize the reconstruction.

C. CS Reconstruction Performance of Uniform Piecewise Extraction

In this part, we consider another extracting mode: uniform piecewise extraction. Here, we give two examples of the extracting matrix Φ : extracting two elements by every six-element intervals and extracting four elements by every eight-element intervals, whose sketch maps are shown in Fig. 9. For the two examples, their respective numbers of measurements are 86 and 128, i.e., the corresponding measurement fractions J/L are 0.34 and 0.50. The test samples, comparable methods (OMP, CVX, and BCS) and their corresponding dictionary setting are same as those in Part B of Section IV. For our method, we also consider two situations: the categories of the training data include those of the test samples ($PM_{1,2,3}$) and the categories of the training data are different from those of the test samples ($PM_{1,2}$, $PM_{1,3}$, and $PM_{2,3}$).

TABLE III

THE MEAN RELATIVE RECONSTRUCTION ERRORS OF 1000 TEST SAMPLE FROM EACH CATEGORY BY OUR METHOD ($PM_{1,2}$, $PM_{1,3}$, $PM_{2,3}$, AND $PM_{1,2,3}$), OMP, CVX, AND BCS WHEN THE EXTRACTING MATRIXES ARE EXTRACTING TWO ELEMENTS BY EVERY SIX-ELEMENT INTERVALS AND EXTRACTING FOUR ELEMENTS BY EVERY EIGHT-ELEMENT INTERVALS

Methods	Extracting two elements by every six-element intervals				Extracting four elements by every eight-element intervals			
	Yak-42 (1000)	Cessna Citation S/II (1000)	An-26 (1000)	Mean (3000)	Yak-42 (1000)	Cessna Citation S/II (1000)	An-26 (1000)	Mean (3000)
$PM_{1,2}$	\	\	0.1542	0.2140	\	\	0.1403	0.1965
$PM_{1,3}$	\	0.1465	\		\	0.1410	\	
$PM_{2,3}$	0.3414	\	\		0.3082	\	\	
$PM_{1,2,3}$	0.2743	0.1161	0.1237	0.1714	0.2567	0.1152	0.1102	0.1607
OMP	0.4999	0.2188	0.2317	0.3168	0.5287	0.2043	0.2907	0.3412
CVX	0.4610	0.2300	0.2016	0.2975	0.4043	0.1746	0.2049	0.2613
BCS	0.4085	0.1897	0.2154	0.2712	0.3762	0.1695	0.1941	0.2466

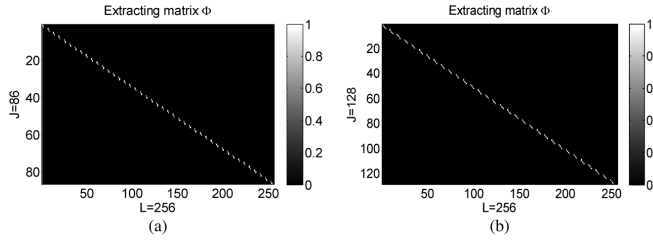


Fig. 9. Two examples of the extracting matrix for uniform piecewise extraction: (a) extracting two elements by every six-element intervals, the corresponding number of measurements is $J = 86$, (b) extracting four elements by every eight-element intervals, the corresponding number of measurements is $J = 128$.

In Table III, we give the mean reconstruction errors of 1000 test samples from each category by our method ($PM_{1,2}$, $PM_{1,3}$, $PM_{2,3}$, and $PM_{1,2,3}$), OMP, CVX, and BCS. From Table III, we can see that the results are similar to those in Table II. Our method, no matter whether the categories of the training data include those of the test samples or not, outperforms the other three approaches for both extracting examples, which demonstrates the effectiveness of the transfer learning and dictionary learning once again. In addition, the reconstruction results of $PM_{1,2,3}$ are still better than $PM_{1,2}$, $PM_{1,3}$, and $PM_{2,3}$, which also indicates the importance of the variety and sufficiency of the training database.

For the purpose of illustration, we show two reconstruction examples in Fig. 10. The reconstructed frequency samples are also transformed to the time domain through IFFT. Here, we still compare our method with OMP, CVX, and BCS. For the first test sample from Yak-42, our method includes $PM_{1,2,3}$ and $PM_{2,3}$, while for the other test sample from An-26, we use $PM_{1,2,3}$ and $PM_{1,2}$. From Fig. 10, we can see that the IFFT HRRP samples reconstructed by our method are accurate enough that the locations of the scattering points are visually approximate to those of the original IFFT HRRP samples. However, the IFFT HRRP samples reconstructed by OMP, CVX, and BCS are less accurate comparing with the original ones, and there are some spurious scattering points in the IFFT HRRP samples reconstructed by OMP. Therefore, we can conclude that our proposed method is superior to the traditional CS recovery methods for reconstruction. Specially, the reconstructed results in Fig. 10(a) and (c) are almost the same, which means that the training frames from Yak-42 do not help to reconstruct this test sample from Yak-42. In other words, it is verified that there may be a large difference between different frames from the same category for HRR data due to the target-aspect sensitivity.

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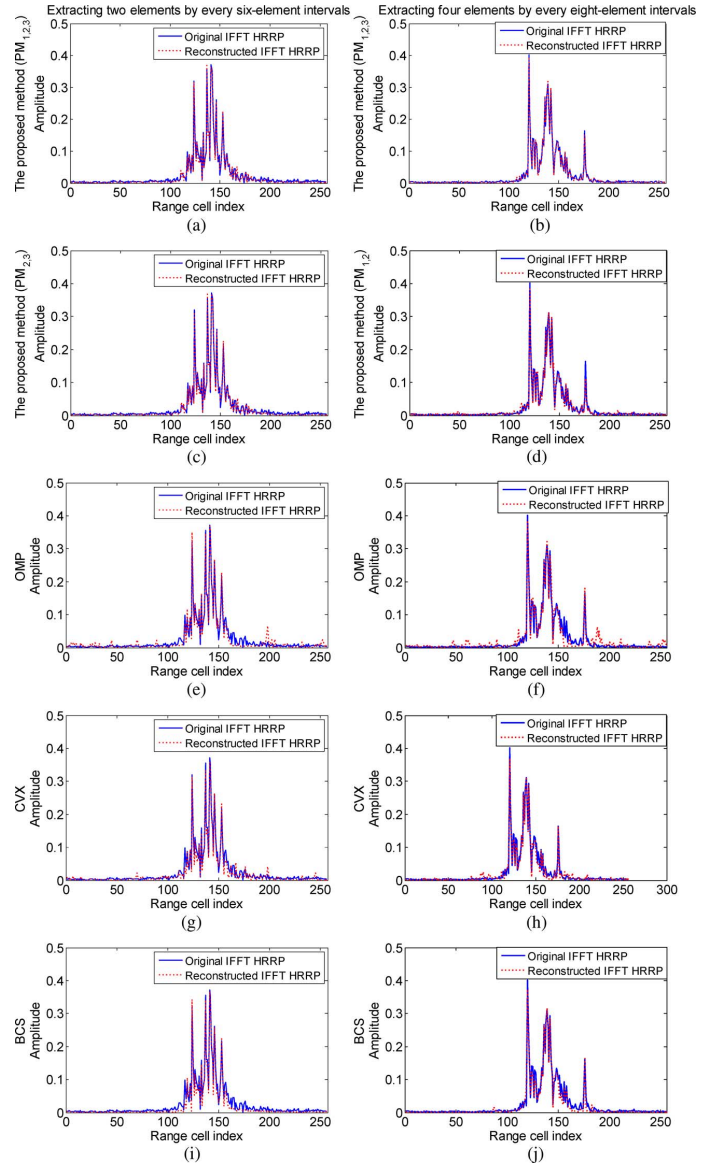


Fig. 10. Reconstruction examples of the HRR data for uniform piecewise extraction. The left column shows one reconstructed IFFT HRRP sample from Yak-42 when the extracting matrix is extracting two elements by every six-element intervals (different methods for each row), and the right column shows another reconstructed IFFT HRRP sample from An-26 when the extracting matrix is extracting four elements by every eight-element intervals. The relative reconstruction errors of the examples are: (a) $PM_{1,2,3}$: 0.1449, (b) $PM_{1,2,3}$: 0.0897, (c) $PM_{2,3}$: 0.1449, (d) $PM_{1,2}$: 0.1429, (e) OMP: 0.2610, (f) OMP: 0.2980, (g) CVX: 0.2249, (h) CVX: 0.2073, (i) BCS: 0.2310, (j) BCS: 0.2069.

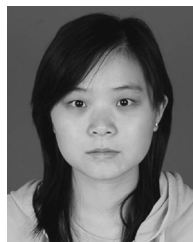
V. CONCLUSION

We develop a new statistical method to realize the reconstruction of the incomplete stepped-frequency data based on transfer learning. With offline training samples, our algorithm can scale to large test data sets at a lower computational cost. We evaluate our method against three traditional CS methods with fixed Fourier dictionary, and the experimental results demonstrate that our method proposes outstanding reconstruction performance.

In the experiments, we mainly show two extracting modes, with each extracting mode only including several examples. We do not discuss which extracting mode can obtain better reconstruction performance. Actually, there is no final conclusion about how to design the projection/extracting matrix Φ so that the measurement is the most informative. Therefore, we are currently investigating different ways to address this problem.

REFERENCES

- [1] D. R. Wehner, *High Resolution Radar*. Norwood, MA, USA: Artech House, 1995.
- [2] B. R. Mahafza, *Radar Systems Analysis and Design Using MATLAB*. Boca Raton, FL, USA: CRC Press, 2000.
- [3] Z. Z. Gao, "New technologies of high resolution ISAR imaging," Ph.D. dissertation, Xi'dian University, Xi'an, China, 2009.
- [4] L. Zhang, Z. J. Qiao, M. D. Xing, Y. C. Li, and Z. Bao, "High-resolution ISAR imaging with sparse stepped-frequency waveforms," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 11, pp. 4630–4651, 2011.
- [5] T. Lamont-Smith, R. D. Hill, S. D. Hayward, G. Yates, and A. Blake, "Filtering approaches for interference suppression in low-frequency SAR," *Proc. Inst. Electr. Eng.—Radar Sonar Navig.*, vol. 153, no. 4, pp. 338–344, 2006.
- [6] E. J. Candès, J. Romberg, and T. Tao, "Robust uncertainty principles: Exact signal reconstruction from highly incomplete frequency information," *IEEE Trans. Inf. Theory*, vol. 52, no. 2, pp. 489–509, 2006.
- [7] D. L. Donoho, "Compressed sensing," *IEEE Trans. Inf. Theory*, vol. 52, no. 4, pp. 1289–1306, 2006.
- [8] R. G. Baraniuk, E. J. Candès, M. Elad, and Y. Ma, "Application of sparse representation and compressive sensing," *Proc. IEEE*, vol. 98, no. 6, pp. 906–909, 2010.
- [9] L. Zhang, M. D. Xing, C. W. Qiu, J. Li, J. L. Sheng, Y. C. Li, and Z. Bao, "Resolution enhancement for inverted synthetic aperture radar imaging under low SNR via improved compressive sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 10, pp. 3824–3838, 2010.
- [10] J. A. Tropp and S. J. Wright, "Computational methods for sparse solution of linear inverse problems," *Proc. IEEE, Special Issue on Applications of Compressive Sensing & Sparse Representation*, vol. 98, no. 6, pp. 948–958, 2010.
- [11] D. Needell and J. A. Tropp, "COSAMP: Iterative signal recovery from incomplete and inaccurate samples," *Appl. Comput. Harmon. Anal.*, vol. 26, no. 3, pp. 301–321, 2009.
- [12] S. H. Ji, Y. Xue, and L. Carin, "Bayesian compressive sensing," *IEEE Trans. Signal Process.*, vol. 56, no. 6, pp. 2346–2356, 2008.
- [13] E. J. Candès and T. Tao, "Near optimal signal recovery from random projections: Universal encoding strategies?," *IEEE Trans. Inf. Theory*, vol. 52, no. 12, pp. 5406–5425, 2006.
- [14] L. Hu, Z. G. Shi, J. X. Zhou, and Q. Fu, "Compressed sensing of complex sinusoids: An approach based on dictionary refinement," *IEEE Trans. Signal Process.*, vol. 60, no. 7, pp. 3809–3822, 2012.
- [15] S. J. Pan and Q. Yang, "A survey on transfer learning," *IEEE Trans. Knowl. Data Eng.*, vol. 22, no. 10, pp. 1345–1359, 2010.
- [16] R. T. Lord and M. R. Inggs, "High resolution SAR processing using stepped-frequencies," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, 1997, vol. 1, pp. 490–492.
- [17] L. Du, H. W. Liu, Z. Bao, and M. D. Xing, "Radar HRRP target recognition based on higher-order spectra," *IEEE Trans. Signal Process.*, vol. 53, no. 7, pp. 2359–2368, 2005.
- [18] L. Du, H. W. Liu, Z. Bao, and J. Y. Zhang, "A two-distribution compounded statistical model for radar HRRP target recognition," *IEEE Trans. Signal Process.*, vol. 54, no. 6, pp. 2226–2238, 2006.
- [19] L. Du, H. W. Liu, and Z. Bao, "Radar HRRP statistical recognition: Parametric model and model selection," *IEEE Trans. Signal Process.*, vol. 56, no. 5, pp. 1931–1944, 2008.
- [20] P. H. Wang, F. Z. Dai, M. Pan, L. Du, and H. W. Liu, "Radar HRRP target recognition in frequency domain based on autoregressive model," in *Proc. IEEE Int. Radar Conf.*, 2011, pp. 714–717.
- [21] S. K. Wong, "Non-cooperative target recognition in the frequency domain," *Proc. Inst. Electr. Eng.—Radar Sonar Navig.*, vol. 151, no. 2, pp. 77–84, 2004.
- [22] W. G. Carrara, R. S. Goodman, and R. M. Majewski, *Spotlight Synthetic Aperture Radar—Signal Processing Algorithms*. Norwood, MA, USA: Artech House, 1995.
- [23] M. Zhou, H. Chen, J. Paisley, L. Ren, L. Li, Z. Xing, D. Dunson, G. Sapiro, and L. Carin, "Nonparametric Bayesian dictionary learning for analysis of noisy and incomplete images," *IEEE Trans. Image Process.*, vol. 21, no. 1, pp. 130–144, 2012.
- [24] M. Aharon, M. Elad, and A. M. Bruckstein, "K-SVD: An algorithm for designing overcomplete dictionaries for sparse representation," *IEEE Trans. Signal Process.*, vol. 54, no. 11, pp. 4311–4322, 2006.
- [25] J. Mairal, F. Bach, J. Ponce, and G. Sapiro, "Online dictionary learning for sparse coding," in *Proc. Int. Conf. Mach. Learn.*, 2009, pp. 689–696.
- [26] J. Paisley and L. Carin, "Nonparametric factor analysis with Beta process priors," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2009, pp. 777–784.
- [27] M. Chen, J. Silva, J. Paisley, C. Wang, D. Dunson, and L. Carin, "Compressive sensing on manifolds using a nonparametric mixture of factor analyzers: Algorithm and performance bounds," *IEEE Trans. Signal Process.*, vol. 58, no. 12, pp. 6140–6155, 2010.
- [28] F. Doshi-Velez, K. T. Miller, J. Van Gael, and Y. W. Teh, "Variational inference for the Indian buffet process," in *Proc. Artif. Intell. Statist.*, 2009, pp. 137–144.
- [29] W. R. Gilks, S. Richardson, and D. J. Spiegelhalter, *Markov Chain Monte Carlo in practice*. London, U.K.: Chapman & Hall, 1996.
- [30] X. H. Ding, L. H. He, and L. Carin, "Bayesian robust principal component analysis," *IEEE Trans. Image Process.*, vol. 20, no. 12, pp. 3419–3430, 2011.
- [31] L. Du, M. H. Chen, J. Lucas, and L. Carin, "Sticky hidden markov modeling of comparative genomic hybridization," *IEEE Trans. Signal Process.*, vol. 58, no. 10, pp. 5353–5368, 2010.
- [32] M. J. Beal, "Variational algorithms for approximate bayesian inference," Ph.D. dissertation, London Univ., London, U.K., 2003.
- [33] L. Du, P. H. Wang, H. W. Liu, M. Pan, F. Chen, and Z. Bao, "Bayesian spatiotemporal multitask learning for radar HRRP target recognition," *IEEE Trans. Signal Process.*, vol. 59, no. 7, pp. 3182–3195, 2011.
- [34] P. H. Wang, "Study of radar high resolution range profile target recognition based on statistical modeling," Ph.D. dissertation, Xi'dian University, Xi'an, China, 2012.
- [35] M. Tipping, "Sparse Bayesian learning and the relevance vector machine," *J. Mach. Learn. Res.*, vol. 1, pp. 211–244, 2001.
- [36] L. Shi, P. H. Wang, H. W. Liu, L. Xu, and Z. Bao, "Radar HRRP statistical recognition with local factor analysis by automatic Bayesian Ying-Yang harmony learning," *IEEE Trans. Signal Process.*, vol. 59, no. 2, pp. 610–617, 2011.
- [37] S. J. Wei, X. L. Zhang, J. Shi, and G. Xiang, "Sparse reconstruction for SAR imaging based on compressed sensing," *Progr. Electromagn. Res.*, vol. 109, pp. 63–81, 2010.



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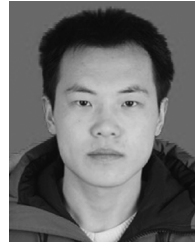


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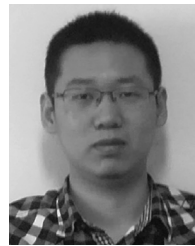
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